

Towards Designing for Multimodal Remembering: Findings from an Interview Study

Ege Otenen
eotenen@iu.edu
Indiana University
Bloomington, IN, USA

ABSTRACT

Humans experience, remember, and express the world through their senses. These different modalities are also essential in forming and recalling autobiographical memories. People remember the color of the shirt they wore, the voice of their partner, and the taste of their mother's soup. They also express their memories with different modalities by telling it with their words or showing pictures. Although a rising number of studies focused on developing technologies to support remembering, studies have yet to take the opportunity to explore the role of modalities to aid memories. We interviewed 12 participants about their memories by allowing them to remember and express memories with different modalities to understand better how to design for multimodal remembering. This paper is a first step toward developing a framework that aims to explore multimodal remembering, improve design practices by providing insights for future memory technologies.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI), HCI design and evaluation methods, Interaction devices.

KEYWORDS

Autobiographical Memory, Sense, Emotion, Memory Technology, Embodied Remembering

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1 INTRODUCTION

Remembering autobiographical memories has received great interest from HCI researchers due to its roots in the sense of self [11] and relation with well-being [25, 30]. Numerous efforts have been made to evoke or aid personal memories [16, 18, 23, 35], designers created technologies to make people remember their personal memories. On the other hand, memory researchers worked immensely to explore the nature of these memories. However, both memory researchers

and designers working to support memories limited their memory collection method to an unimodal channel, mostly verbal or written. Even though scholars examined the role of different modalities on remembering a memory by comparing memories prompted with visual, olfactory, auditory cues [10, 14, 17, 21, 29, 38, 47] or aiding the memories with the display of visual [23, 42], olfactory components [2], their focus was mainly in the “remembering” phase of the memory. However, the second phase, where people express their memory, which is the focus of interest while conducting a study, was almost always limited to verbal descriptions [8, 16, 41]. All the sensory-perceptual knowledge a memory has was inducted to one dimension and reflected only with words. The multimodal nature of memory traces evoked in the remembering phase has been neglected (see [24] for a review). Due to the multimodal structure of our memories, which is limited to unimodal expression, the current study aims to explore the influence of focusing on different modalities of autobiographical memories on recalling and expressing them.

2 RELATED WORK

2.1 HCI and Autobiographical Memory

Previous technologies mainly focused on visual modality to aid memory. For instance, the tools were mainly designed to facilitate the life-logging via taking pictures (e.g., SenseCam [23], AffectCam [42], Narrative Clip [48], MyLifeBits [19]). These life-logging technologies aim to strengthen memory with a mini camera that takes random pictures at different intervals in a day, carried by people who then review these pictures. Some of these technologies also encourage people to reflect on these pictures [41] with pop-quizzes, survey-like questions [16] or verbally summarize the content they see in their life logs [16]. These reflections and summaries boost individuals' memory by helping them to reminisce and share it with others [32]. With technological improvements, people are not limited to remembering the past via reviewing pictures or engaging with verbal questions about their memories. They also benefit from different methods targeting different modalities. For instance, Dib et al.[15] demonstrated that listening to recorded sounds during a family holiday creates a more unique and special space for the family to reminisce about the holiday more creatively and naturally than reviewing holiday pictures. Another modality, olfaction also improves memory, so that participants who are exposed to olfactory stimulation during the memorization task performed better than the control group receiving no olfactory cue [2].

2.2 Memory and Modalities

Like HCI, memory studies also almost always relied on the verbal or written description of memories [33, 43], so that when people

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are asked to tell their memories, they either verbally describe them or write them down. Even the studies focused on multimodal nature of memories only examined the role of modality regarding their impact in triggering the memories [10, 14, 17, 21, 29, 38, 47]. Researchers used different unimodal or multimodal cues (e.g., visual, auditory, olfactory) to prompt memories. Still, they asked people to describe them verbally [47]. No study examined the role of modalities to express or describe the memories. Autobiographical memories, characterized as specific events happening to the self in a particular place and time, on the other hand, heavily rely on sensory-perceptual information [12]. Indeed, sensory-perceptual knowledge is the main component of memories distinguishing a memory from imagination [26]. Autobiographical memories consist of information about the place, people, perceptual information, time, emotions, and thoughts, and having this sensory-perceptual information is essential for the richness of the memories [33]. Given the role of perception and indirect use of senses, it is surprising that no study or design has examined the influence of having multimodal memory expressing opportunity on memory. Considering the embodied cognition literature suggesting that the body supports memory either by lowering the cognitive load with its movements (e.g., gestures) [45], or imitating the previous movements back in the memory [24], and the recent findings about the relation between body movements (gestures) and telling episodically rich autobiographical memories [5]; embodiment is becoming more interesting for memory technologies. HCI researchers are trying to understand how to design for embodied remembering. For example, Overvest [36] conducted a diary study examining the role of body use on remembering. It is evident that visual, auditory, olfactory, and tactile components support memory and multiple designs were proposed using these modalities. Although these different modalities were used in technology design to aid memory, very few focused on a multimodal design that combines a couple of these modalities. Only Kim et al. (2022) [27] designed Slide2Remember; a tool for reminiscing the past with visual, auditory, and tactile experiences in which people reminisce their memory by seeing the picture of it hanged in their place and physically interacting with it by sliding the frame of the picture and listening a song related to the picture. Considering the role of different modalities on memories, there needs to be more designs that focus on using multimodal components to aid the recall and expression of memories.

2.3 Current Study

This study aims to understand the value of modalities in remembering and expressing autobiographical memories by asking the following questions:

- 1- Do remembering with different modalities support/contribute to autobiographical memory?
- 2- How is the experience of remembering and expressing the memory with modalities?

3 METHODOLOGY

Interviews were conducted with 12 Turkish adults (5 male, seven female). The number of participants has been decided based on the previous qualitative memory studies in HCI [13, 18, 46]. This

number has been reported in a review paper as the most common sample size in qualitative studies in ACM CHI conference [9]. Thus, it should suffice to understand the general effect of modality on memories. Most participants ($N = 10$) were young adults (25–34 years old); one was in the range of 35 to 44 years old, and one was older than 55. They were recruited with convenience sampling.

4 PROCEDURE

Participants were contacted to set a convenient meeting place and time. Then, they met with the experimenter in person in a quiet space. After agreeing on informed consent, participants were instructed to remember recent and emotional memories and express the memory with a memory board while the session was audio-recorded. The main focus was on recent memories because they are easily accessible and their sensory features are preserved [44], along with emotional memories which has higher level of sensory features and they are easily remembered with a greater detail than neutral memories [43]. Also having both emotional and mundane memories can provide insights into the interaction between memories and modalities. Following the memory part, they completed a couple of survey questions about the characteristics of their memory (e.g., vividness, emotion, time of the event [39]), and they repeated the same steps for the second memory. Lastly, participants completed an individual differences survey on their well-being [40], imagery [7] and representation skills [37], and demographics, yet these will not be discussed within the scope of this paper.

4.0.1 Step 1. Remembering a Memory. Half of the participants started with thinking about a recent autobiographical memory of them, while the other half started with an emotional one. With adapted instructions [1, 4] participants were asked to remember memories lasting no longer than a day and happening to them in a specific place and time.

4.0.2 Step 2. Expressing a Memory. All participants were provided with crayons, play dough, and blank paper, and a music player. The following instruction was given to them:

"Let's name this experience as memory board. I want you to remember/describe your memory as your own way with anything that makes sense to you and feels right. You are free to use any sensations coming to you and reflect on them, you can create visuals (e.g., diagrams, sketches), write down things, attach a smell, taste, heat, pain, balance, sound, body movement, or any texture to the memory. Please use any material/equipment you want to describe your memory. If you cannot include any modality due to plausibility, feel free to write it down in your memory board. Take as much as time you need to re-create your memory."

4.0.3 Step 3. Matching Modalities with Memory. After the expression of memory, participants were provided with ten modality cards: Visual, Auditory, Gustatory, Olfactory, Tactile, Kinesthetics-Proprioception (sensing the position of body or body parts), Kinesthetics-Equilibrium (sensing balance), Haptic (sensing any external touch passively). These modalities were selected from Augstein and Thomas Neumayr's work [3] that considers the human-computer interaction when categorizing modalities. Thermoception (sensing heat and temperature) and Nociception (sensing physical pain) were added from the somatosensory category [22]

to cover the sensory experience. Participants were first provided with the modality explanations and then asked to match them with memory details if they existed (e.g., "Jingle bells was playing" → Auditory).

4.0.4 Step 4. Reflections on Modalities and Memory Board. Upon completing modality memory matching, participants answered if there were any modalities they would prefer to express their memory and reported the best-expressing modalities. Lastly, they reflected on the memory board experience.

5 PRELIMINARY FINDINGS

Interviews revealed some critical points about the role of modalities in remembering and expressing memories. First, I describe the content of memories and then report how many modalities are used in the remembering and expression phases. A total of 24 memories (12 recent and 12 emotional) were collected from 12 participants. Six of the emotional and ten of the recent memories were positive, while the rest (six emotional, two recent) were negative. Table 1 demonstrates themes observed in memories and their counts, they were common themes observed in autobiographical memory research [6].

5.1 Modalities

5.1.1 Modalities Matching with Memory Details. When participants were asked to match the modalities with the details in their memories, the most common modality was visual; all memories had visual details except one. Although all modalities were included in at least one memory; auditory, olfactory, tactile and gustatory modalities were commonly observed in almost all memories. Table 2 and Table 3. shows the number of memories, including these modalities. Both recent and emotional memories displayed similar patterns regarding the frequency of modalities.

Memories	#
Specific conversation with someone	6
Trip / going out	6
Accident	3
Birth of someone	2
Therapy session	1
Mistake made in workplace	1
Funny event in childhood	1
Loss or sickness of a loved one	1
Theatre scene	1
Special event (e.g., wedding)	1

Table 1: Memory Themes and Counts.

5.1.2 Best Representing Modalities. When participants were asked about the modalities that can represent their memory the best, the frequency of the modalities slightly changed. Overall, visual, tactile, and auditory were the best modalities. See Table 4 and Table 5 for modalities and counts of memories. They were picked as the best modalities. Also, for most of the memories (14 out of 24), people reported they prefer a multimodal representation, which means they believe their memory would be represented the best with

Emotional Memories	#
Visual	12
Auditory	10
Olfactory	7
Tactile	7
Gustatory	5
Haptic	5
Thermoception	4
Kinesthetics-Equilibrium	5
Kinesthetics-Proprioception	4
Nociception	3

Table 2: Emotional Memory Detail Modalities and Counts.

Recent Memories	#
Visual	11
Auditory	10
Olfactory	8
Tactile	8
Gustatory	8
Haptic	3
Thermoception	6
Kinesthetics-Equilibrium	5
Kinesthetics-Proprioception	4
Nociception	2

Table 3: Recent Memory Detail Modalities and Counts.

Emotional Memories	#
Visual	8
Auditory	4
Olfactory	0
Tactile	7
Gustatory	2
Haptic	1
Thermoception	1
Kinesthetics-Equilibrium	0
Kinesthetics-Proprioception	2
Nociception	1

Table 4: Best Representing Modalities and Counts - Emotional Memories.

more than one modality. These findings show that best representing modalities (visual, tactile, and auditory) are not necessarily the ones salient in the memories (e.g., olfactory, gustatory).

5.1.3 Modalities for Expressing Memory. All participants verbally described their memories. Indeed, four memories were described only verbally. Participants also preferred sketching (19 out of 24 memories). The main themes observed in these drawings were the spatial structures of the places (e.g., house, arrangement of chairs) and the components of the event (e.g., objects, people). The first one was unsurprising since scene construction is a critical process in memory constructions [34]. Twelve of the drawings include more than one color. Participants also wrote down words (12 out

Recent Memories	#
Visual	10
Auditory	4
Olfactory	2
Tactile	4
Gustatory	2
Haptic	0
Thermoception	1
Kinesthetics-Equilibrioception	1
Kinesthetics-Proprioception	1
Nociception	1

Table 5: Best Representing Modalities and Counts - Recent Memories.

of 24 memories) and used numbers in two memories. Participants used physical objects to imitate the bodily actions happening in their memories. One described the sitting position with a chair, and the other used a pen as the knife. Furthermore, two participants played songs they were listening to in the memory; one played a song representing the memory. The dough was used as a tool and in the drawings (six memories). Interestingly, all participants except the four played with dough during the memory expression. Considering the positive relation between gestures and memory details [5], playing with dough might facilitate memory expression by the movement of hands.

5.1.4 Implausible Modalities for Expressing Memory. When participants were asked what would represent their memories but not plausible, they reported photos, videos, voice recordings, smells, objects, and human figures would represent the best. Also, three participants stated the emotion of the memory would represent it well.

5.2 User Experience for Memory Board

Most participants recommended the memory board since they felt good, happy, and excited about it. Considering the positive influence of remembering positive memories on mood [20], it was not surprising. To understand the separated effect of the memory board from remembering, participants were asked if it would feel the same when they told the memory with words only. They said the memory board created an awareness in their mind about the other aspects of their memory by providing different perspectives, making their memory more concrete. Thus, it was a different experience than merely telling the memory.

5.2.1 Memory Board Helps to Remember and Express. Participants stated the memory board helped them remember their memories better than expected. Remembering with memory board increased memory's intensity and reality, making it tangible. They felt like reliving it again. Some participants also stated they felt like they expressed their memory better because the memory board provided room to be creative compared to verbal descriptions. It enabled people to share nonverbal features. For instance (see Figure 1), Participant 109 indicated that adding light particles as dough pieces on top of the table helped her emphasize the importance of light in this memory, and she would not describe it with words. Some

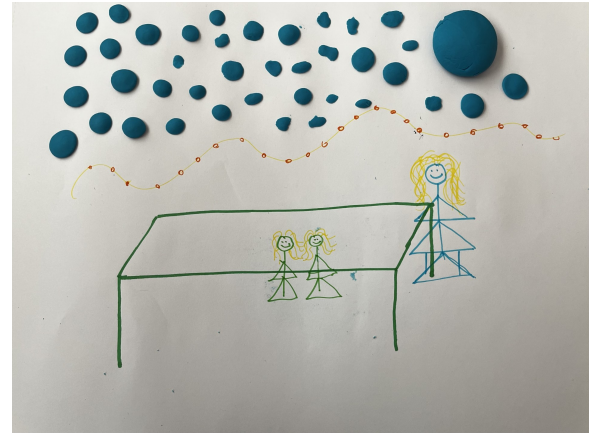


Figure 1: Memory Board - Participant 109

participants indicated that the memory board influenced their remembering, some said their expression, and some argued it is both. There were also unfavorable effects of it on memory. For instance, Participant 111 thought the memory board only influenced remembering; he stated recalling his friend sitting next to him, but because this friend was telling a story, he pictured him as he was sitting across him -as the story-teller of the memory-, and that's why he drew the friend sitting right across him in the memory board even though it was not his seat.

5.2.2 Memory Board Helps to Create. A couple of participants discussed the room memory board provided to create and reflect on their memory. For instance, Participant 109 described a fun memory she thought her friend resembled a fictional character from a movie, and she drew her by adding more visual cues to make her look like that character. Therefore, rather than accurately depicting the moment, she exaggerated the humorous parts in her sketchings. Another pattern observed is that participants drew symbols and figures related to concepts. Participant 104's memory had distinct cinnamon and pumpkin smells, and he drew pumpkin faces and Halloween costumes even though these concepts were not in the memory itself. Participant 103 also preferred to draw hearts to the memory to reflect the event's emotion. Another interesting finding was about the visuals drawn on the memory board. Two participants drew non-existing objects that they visually imagined within their memory. For instance, Participant 106 was describing a memory of getting bitten by a donkey and immediately thinking about the scary vaccine she needed to get. Even though the needle was in her imagination, she drew a needle on the memory board. Overall, the memory board facilitated individuals' creative reflection by allowing them to draw connections between their memory, thoughts, and imaginations.

5.2.3 Memory Board is Useless for Negative Memories. Not all participants suggested the memory board experience. Three participants recalling negative memories found the memory board redundant or painful due to the increased intensity of negative emotions. For instance, Participant 111 refused to draw anything since he felt it would hurt to visualize that memory. Attitude toward the

memory board may be due to participants' emotion regulation strategies. Since the memory board leads to a self-immersed experience, participants might refuse to do it to reduce negative emotions by distancing them from memory. It is known that self-distancing reduces the negative emotions [31]. This finding alone highlights the role of modalities to support memory because multimodal memory board experience leads to reliving the memory very strongly.

6 CONCLUSION AND DISCUSSION

This paper is the first to examine the contributions and experience of remembering and expressing memories with modalities. The findings show the functions of the memory board, especially for positive memories, and highlight the discrepancy between the modalities existing in memories and the ones picked as best representing. Moreover, it tests the user experience of the memory board, which was overall indicated as a positive experience since it aids remembering, facilitates memory expression, and helps people reflect on their memories creatively. However, these reflections were specific to positive memories. Future designs should consider the emotion of the memories and the aim of users about the memory when implementing the modalities into their designs.

This research might inform scholars, especially those designing memory-aiding technologies, due to the insights it provides. The first important point this study hints at is that tactile is a modality that exists and is preferred in memory recalls and representations but lacks in current technologies. Considering the improvement the tactile interaction with daily objects creates in dementia patients [28], memory technologies should focus more on implementing tactile interactions. Second, people mainly use verbal descriptions, visuals (e.g., sketching and coloring), and written words to express their memories. They believe some implausible components: visual (e.g., photo, video), auditory (e.g., video, voice), olfactory (e.g., smell), embodied (e.g., sitting in the chair), and tactile modalities (e.g., human figures as toys, soft objects) would describe their memories best. It signals a gap in the memory technologies to include olfactory and embodied representations in addition to tactile ones since participants think they would represent the memories but could not include them due to plausibility. Last, emotion has a significant role in expressing the memory. The relation between emotion and modalities can be investigated thoroughly for future designs, and ways to make emotion tangible can be explored.

In addition to the unimodal inclusion of modalities, this study showed the prevalence of multimodal experience. Future designs should explore the possible implementation of multimodal systems. Another finding is the absence of any difference in recent and emotional memories regarding the number of modalities and best-representing modalities observed. It might be showing individual preferences or the skills to remember and represent the memory, independent from the content (e.g., being emotional) or the time of the memory (e.g., recent). There is a need for understanding personas and individualized memory technologies for them.

In the next steps of this study, I aim to extract personas based on the participants' visual imagery and mental representation skills and explore whether their abilities align with their modality preferences. With the knowledge obtained through this research, I aim to build a framework for designing multimodal memory-aiding

systems, which will be a novel contribution to the HCI field. I desire this project to raise awareness of a neglected design domain: multimodal remembering.

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